CHAPTER 18

Saving, Investment, and the Financial System

ESSENTIALS OF
Economics
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Premium PowerPoint Slides
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Financial Institutions

- The **financial system**: the group of institutions that helps match the saving of one person with the investment of another.
- **Financial markets**: institutions through which savers can directly provide funds to borrowers. Examples:
  - The Bond Market.
    A **bond** is a certificate of indebtedness ← Debt finance.
  - The Stock Market.
    A **stock** is a claim to partial ownership in a firm. ← equity finance.

Financial Institutions

- **Financial intermediaries**: institutions through which savers can <u>indirectly</u> provide funds to borrowers. Examples:
  - Banks: Medium of exchange beyond just store of value
  - **Mutual funds** – institutions that sell shares to the public and use the proceeds to buy portfolios of stocks and bonds

Different Kinds of Saving

Private saving

= The portion of households' income that is not used for consumption or paying taxes

= **Y – T – C**

Public saving

= Tax revenue less government spending

= **T – G**

National Saving

National saving

= private saving + public saving

$$= (\mathbf{Y} - \mathbf{T} - \mathbf{C}) + (\mathbf{T} - \mathbf{G})$$

$$= \mathbf{Y} - \mathbf{C} - \mathbf{G}$$

= the portion of national income that is not used for consumption or government purchases

Saving and Investment

Recall the national income accounting identity:

$$\mathbf{Y} = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

For the rest of this chapter, focus on the closed economy case:

$$\mathbf{Y} = \mathbf{C} + \mathbf{I} + \mathbf{G}$$

Solve for **I**:

$$\mathbf{I} = \mathbf{Y} - \mathbf{C} - \mathbf{G} = \overbrace{(\mathbf{Y} - \mathbf{T} - \mathbf{C}) + (\mathbf{T} - \mathbf{G})}^{\text{national saving}}$$

Saving = investment in a closed economy

Budget Deficits and Surpluses

Budget surplus

= an excess of tax revenue over govt spending

= **T – G**

= public saving

Budget deficit

= a shortfall of tax revenue from govt spending

= **G – T**

= – (public saving)

ACTIVE LEARNING 1

A. Calculations

- Suppose GDP equals $10 trillion, consumption equals $6.5 trillion, the government spends $2 trillion and has a budget deficit of $300 billion.
- Find public saving, taxes, private saving, national saving, and investment.

ACTIVE LEARNING 1
Answers, part A

Given:

Y = 10.0, **C** = 6.5, **G** = 2.0, **G** – **T** = 0.3

Public saving = **T** – **G** = – 0.3

Taxes: **T** = **G** – 0.3 = 1.7

Private saving = **Y** – **T** – **C** = 10 – 1.7 – 6.5 = 1.8

National saving = **Y** – **C** – **G** = 10 – 6.5 = 2 = 1.5

Investment = national saving = 1.5

ACTIVE LEARNING *1*

B. How a tax cut affects saving

- Use the numbers from the preceding exercise, but suppose now that the government cuts taxes by $200 billion.
- In each of the following two scenarios, determine what happens to public saving, private saving, national saving, and investment.
  1. Consumers save the full proceeds of the tax cut.
  2. Consumers save 1/4 of the tax cut and spend the other 3/4.

ACTIVE LEARNING 1
Answers, part B

In both scenarios, public saving falls by $200 billion, and the budget deficit rises from $300 billion to $500 billion.

1. If consumers save the full $200 billion, national saving is unchanged, so investment is unchanged.

2. If consumers save $50 billion and spend $150 billion, then national saving and investment each fall by $150 billion.

ACTIVE LEARNING *1*

C. Discussion questions

The two scenarios from this exercise were:

1. Consumers save the full proceeds of the tax cut.
2. Consumers save 1/4 of the tax cut and spend the other 3/4.

- Which of these two scenarios do you think is more realistic?
- Why is this question important?

The Meaning of Saving and Investment

- **Private saving** is the income remaining after households pay their taxes and pay for consumption.
- Examples of what households do with saving:
  - Buy corporate bonds or equities
  - Purchase a certificate of deposit at the bank
  - Buy shares of a mutual fund
  - Let accumulate in saving or checking accounts

The Meaning of Saving and Investment

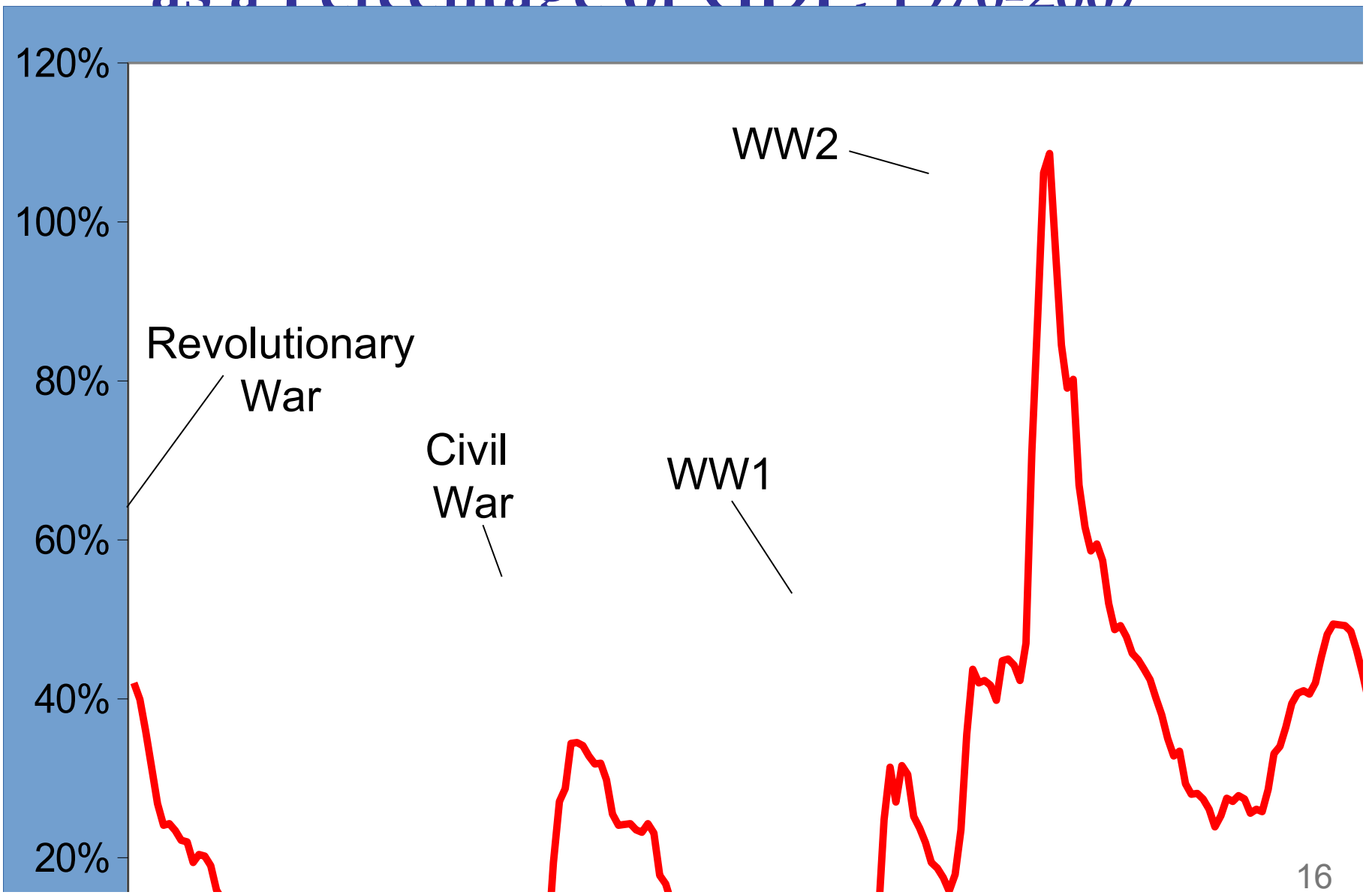
- **Investment** is the purchase of new capital.
- Examples of investment:
  - General Motors spends $250 million to build a new factory in Flint, Michigan.
  - You buy $5000 worth of computer equipment for your business.
  - Your parents spend $300,000 to have a new house built.

Remember: In economics, investment is NOT the purchase of stocks and bonds!

The U.S. Government Debt

- The government finances deficits by borrowing (selling government bonds).
- Persistent deficits lead to a rising govt debt.
- The ratio of govt debt to GDP is a useful measure of the government's indebtedness relative to its ability to raise tax revenue.
- Historically, the debt-GDP ratio usually rises during wartime and falls during peacetime – until the early 1980s.

U.S. Government Debt as a Percentage of GDP, 1970-2007



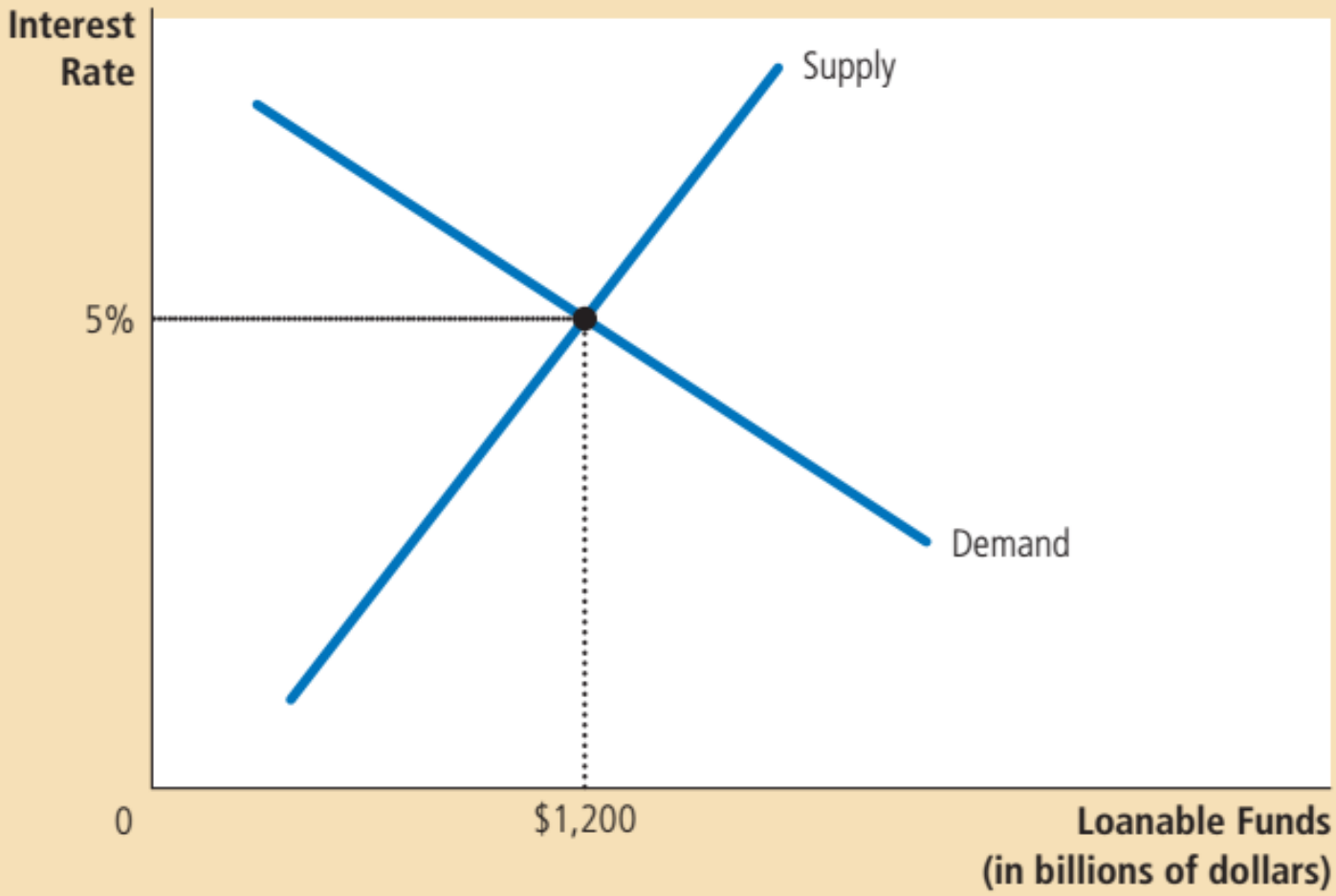
Market for Loanable Funds

- Just market for demand and supply for borrowing and saving
- Loanable funds:
  - income people have chosen to save and lend out
  - Or amount investors have chosen to borrow for new investment

What economic variable captures the price of demand for funds and also price of supply of funds?

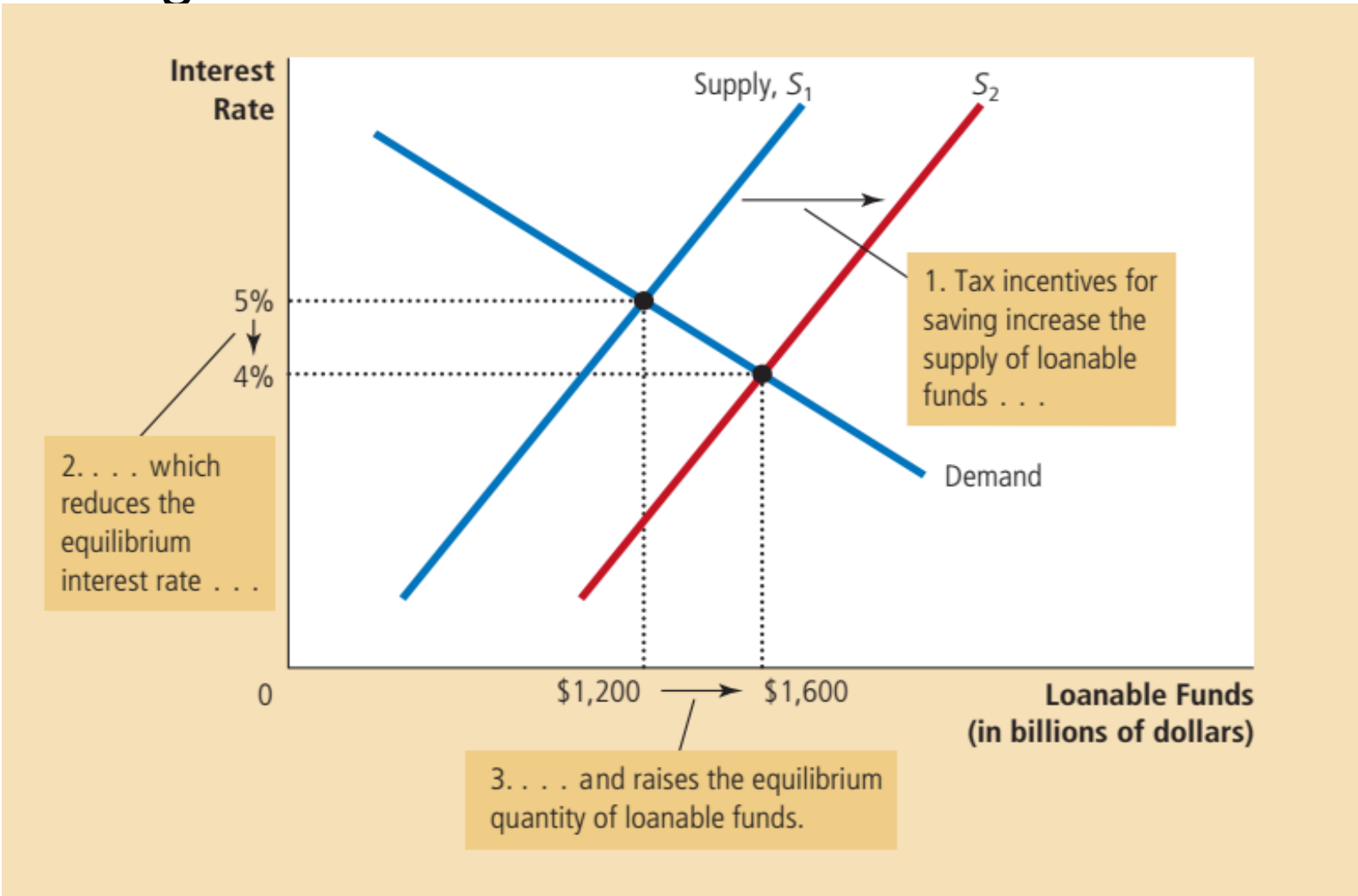
- Return to savings and cost of borrowing

- Source of supply of loanable funds ← savings
- Source of demand of loanable funds ← investment


Interest Rate
Supply
5%
Demand
0
$1,200
Loanable Funds
(in billions of dollars)

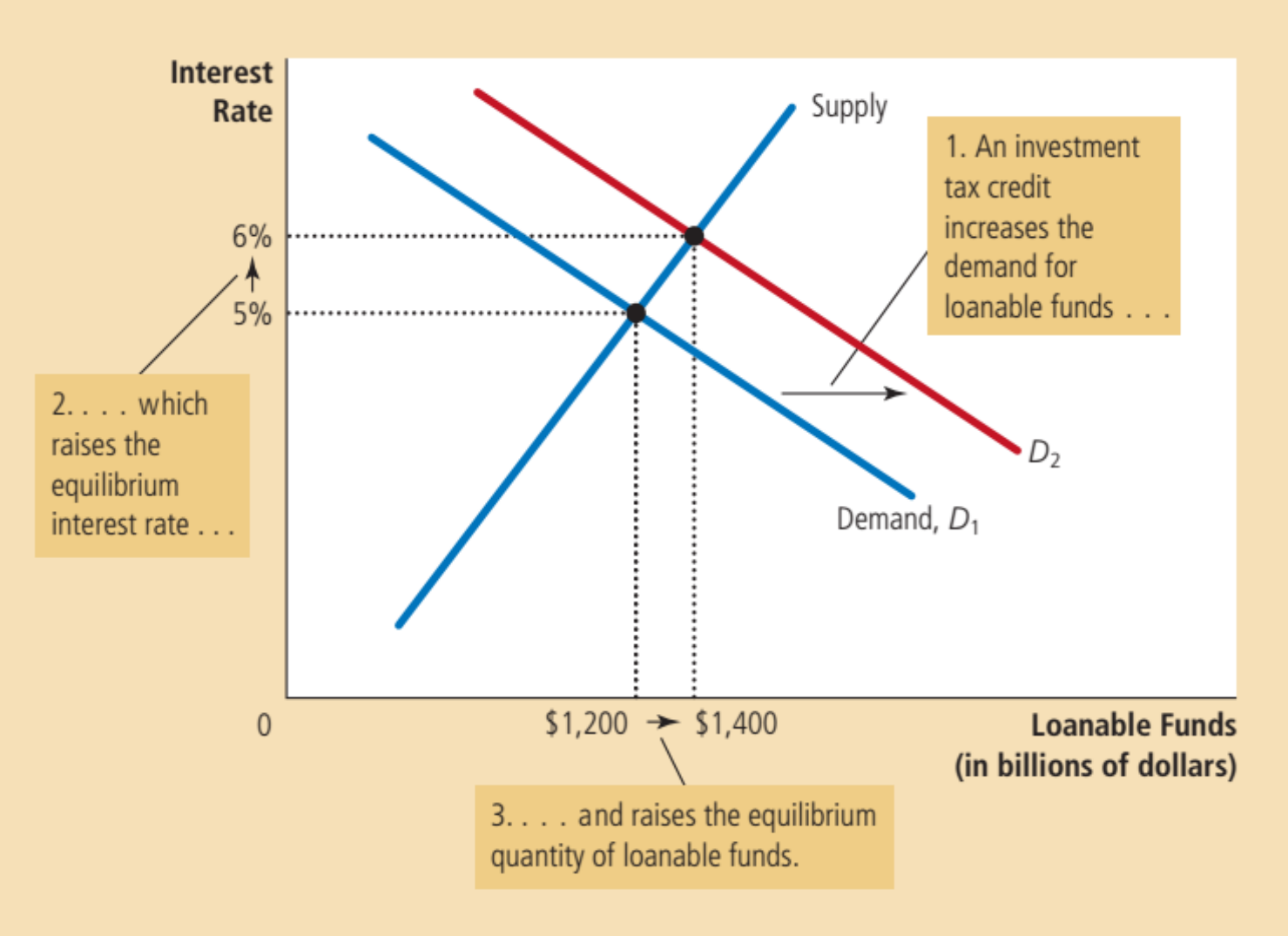
How Policies affect this Market

- Savings Incentives?



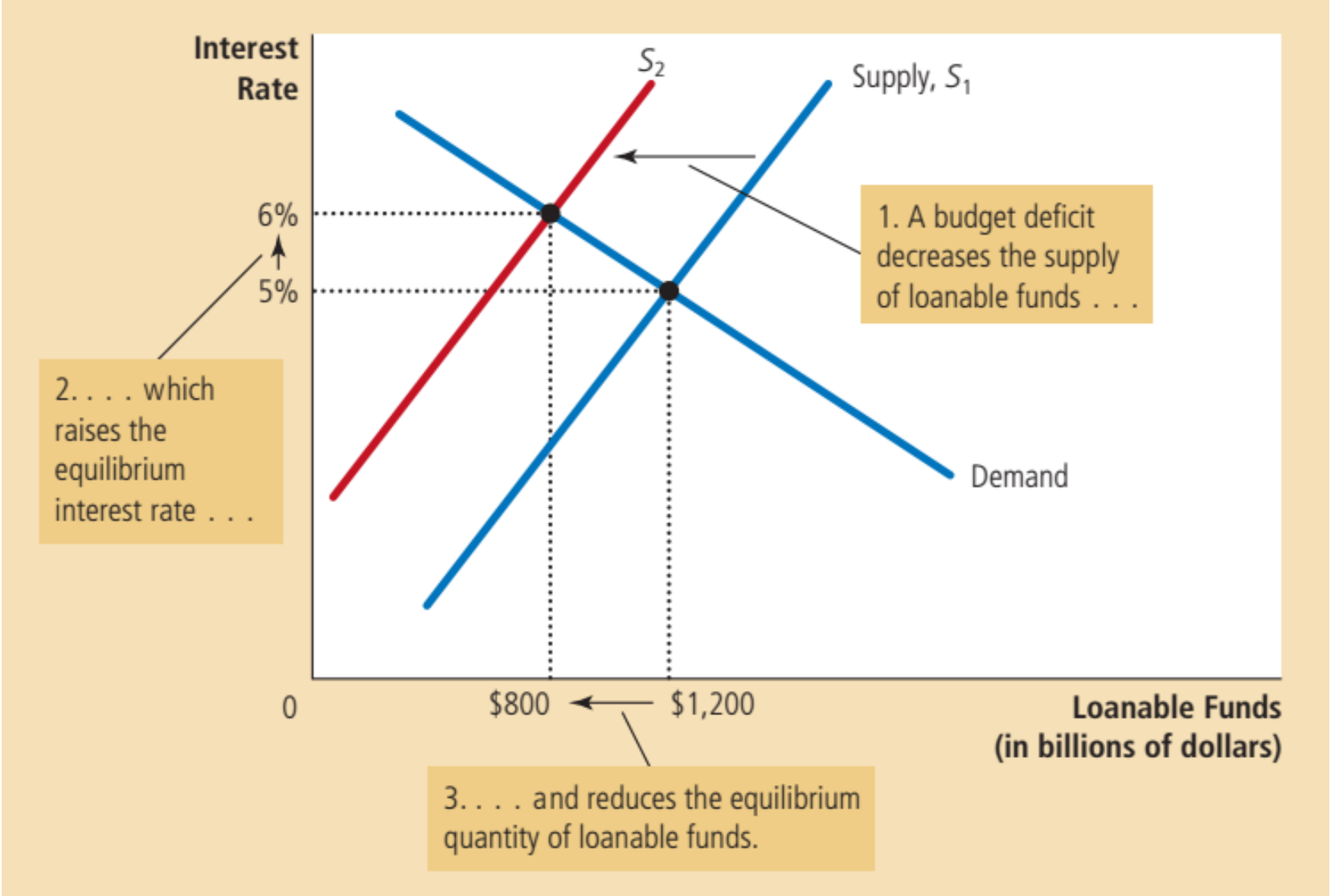
How Policies affect this Market?

- Investment Incentives



How Policies affect this Market?

- Government Budget Deficits and Surpluses
  - Budget deficit means greater spending by government than matched by its income
  - Hence greater demand for loanable funds by the government itself
  - *Government borrows from the bond market or stock market?*
  - Supply of loanble funds comes from national savings
  - Deficits reduce public savings to negative
  - Supply curve shifts backwards → **crowding out effect**
  - Why dont we shift demand curve forwards?


Interest Rate
S_2
Supply, S_1
6%
5%
1. A budget deficit decreases the supply of loanable funds . . .
2. . . . which raises the equilibrium interest rate . . .
Demand
0
$800
$1,200
Loanable Funds (in billions of dollars)
3. . . . and reduces the equilibrium quantity of loanable funds.